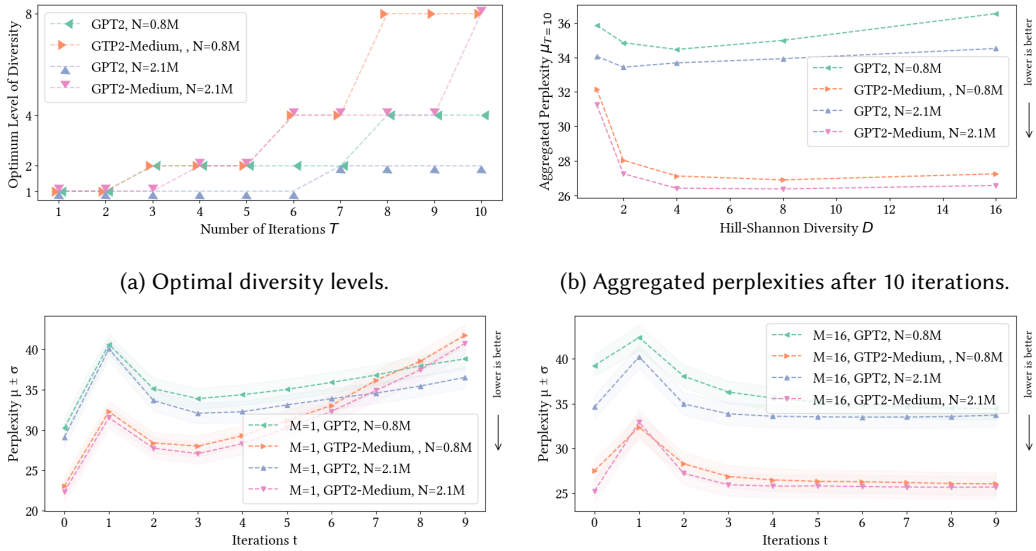


(a) Optimal diversity levels.

(b) Aggregated perplexities after 10 iterations.

(c) Mean perplexities for M;D=1

(d) Mean perplexities for M;D=16

Fig. 14. **The optimal level of diversity for GPT-2 performance increases as both model and dataset size grow.** Ecosystem diversity across four system scales (two model sizes × two dataset sizes; variation V1). "GPT-2, $N = 0.8$M" (green ◀) corresponds to the primary setting, using a 124M-parameter model and a total training dataset of 0.8M tokens. "GPT-2 Medium" corresponds to the 355M-parameter model in the GPT-2 family. **Top:** (a) Optimal diversity level vs. number of iterations T and (b) aggregated perplexity $\mu_{T=10}$ for five diversity levels ($M; D = 1, 2, 4, 8, 16$). Larger models (orange and pink vs. green and blue) show a steeper increase in optimal diversity with T, shifting the optimum from $D = 4$ to $D = 16$ after 10 iterations. At a fixed model size, increasing dataset size (pink ▼ and blue ▲ vs. green and orange) decreases the optimal level. **Bottom:** Perplexity trajectories across ten iterations for two diversity levels (c: $D = 1$, d: $D = 16$). For a fixed diversity level, increasing model or dataset size reduces perplexity at $t = 0$. (c) In the most homogeneous ecosystem ($D = 1$), larger models show strong perplexity increases in later iterations, indicating stronger collapse, while solely increasing dataset size slightly decreases perplexity. (d) In highly diverse ecosystems ($D = 16$), increasing model and/or dataset size decreases perplexity. The shaded region represents the combined standard deviation across models and data points.

Hyperparameter	Value
temperature	0.0
do_sample	False
max_new_tokens	128
min_new_tokens	128
repetition_penalty	0.0
batch size	32
block size	128
num_beams	5

Table 1. Generation hyperparameters.

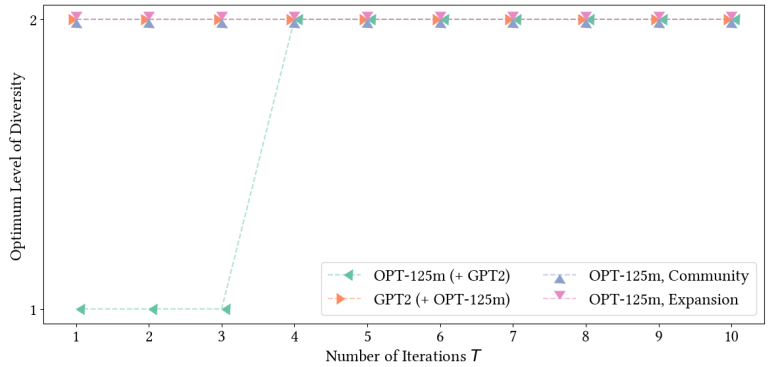
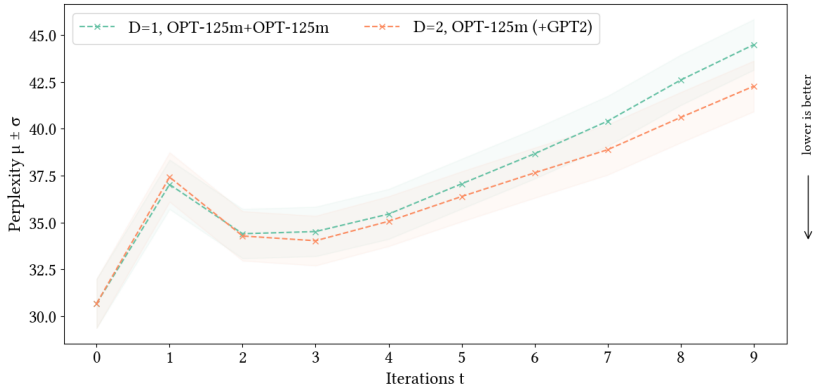
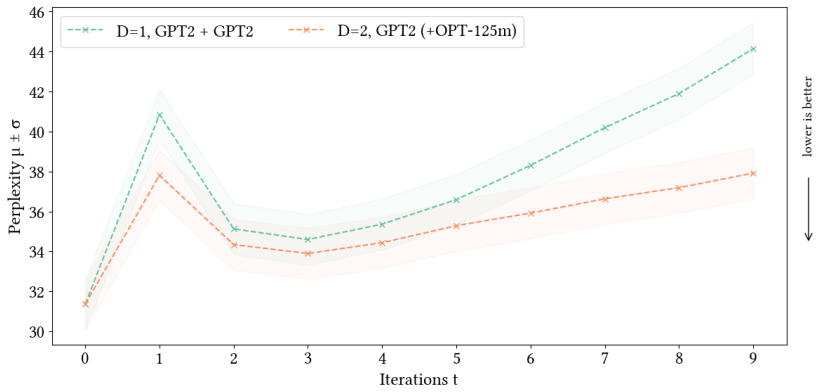


Fig. 15. **The advantage of homogeneous ecosystems observed short-term can even disappear entirely under three modeling variations:** introducing diversity by combining model families rather than segmenting data ("OPT-125m(+GPT2)" and "GPT2(+OPT-125m)") instantiating diversity via dataset expansion rather than dataset segmentation ("OPT-125m, Expansion"), and evaluating community models on community-specific test sets rather than a universal benchmark ("OPT-125m, Community"). These variations involved only two diversity levels ($D = 1, 2$).



(a) Combination of two OPT-125m models compared with an OPT-125m paired with GPT-2



(b) Combination of two GPT-2 models compared with an GPT-2 paired with OPT-125m

Fig. 16. **Two diverse pre-trained models yield lower perplexity than two identical pre-trained models.** OPT-125m+OPT-125m and GPT2+GPT2 represent ecosystems composed of two identical models (green). OPT-125m(+GPT2) and GPT2(+OPT-125m) represent ecosystems where each model is paired with the other type (orange). Unlike the primary segmentation approach, diversity here is instantiated by using two different pre-trained models. All models are trained on the same dataset at iteration step $t = 0$ and use the same generation dataset throughout; thus, aside from their pre-training differences, the training procedure is identical across models. The shaded region represents the combined standard deviation across models and data points.